

Reproducing a Physics-Informed Pedestrian Dead-Reckoning Study on Public Data with PINNeAPLe

EKF with velocity-predicting networks (adapted TLIO, DIVE) and grid-output PINN variants, after Ligthart (2025)

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PINNeAPLe Labs — Physics AI research notes

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Abstract

Ligthart's MSc thesis [1] studied a pedestrian dead-reckoning (PDR) algorithm in which an extended Kalman filter (EKF) is corrected by a neural network that predicts walking velocity [2], and asked whether physics-informed training improves it. It concluded that an adapted TLIO network beats DIVE, that a bias in the predicted velocity is the bottleneck, and that a physics loss does not help. The thesis used a proprietary 25-hour dataset; we repeat the study with PINNeAPLe on the public RIDI dataset (2.7 h, person-disjoint split). Adapted TLIO again outperforms DIVE: median 90-second drift 4.5 % vs 7.8 % (paired Wilcoxon $p = 0.0053$). The best network's error has a significant mean in the heading frame (-0.084 m/s along and -0.036 m/s across the walking direction, $|z|$ up to 51), confirming the bias the thesis identified. A physics loss from the accelerometer model did not significantly improve drift for $\lambda \in \{0.01, 0.1, 1.0\}$ relative to $\lambda = 0$ (Bonferroni-corrected), consistent with the thesis.

Keywords: pedestrian dead reckoning; inertial navigation; extended Kalman filter; PINN; TLIO; reproducibility; PINNeAPLe

What differs from the thesis

- Data: public RIDI smartphone data (2.7 h, 11 people) instead of Leica's proprietary 25 h; absolute numbers are not comparable, only the direction of each conclusion.
- Filter: state (position, velocity) with the phone's own orientation; the thesis' EKF also estimated attitude and IMU biases.
- Drift is measured over fixed 90 s segments of the test sequences, not over whole experiments.
- Networks: PINNeAPLe Conv1DModel (dilated residual 1-D CNN) backbones instead of the thesis' ResNet/U-Net/Transformer variants.

1 Background

In PDR the IMU is integrated forward and the drift that accumulates from sensor errors is corrected by a pseudo-measurement. In the algorithm of Liu et al. [2] (TLIO) the pseudo-measurement is a velocity (or displacement) with its uncertainty, predicted by a network from a window of IMU data. The EKF assumes that this measurement error has zero mean; the thesis showed that the networks violate this and that the violation limits accuracy. DIVE [3] instead rotates the IMU data into a gravity-aligned, yaw-free frame.

2 Methods

- Data: RIDI [4], 94 smartphone sequences with Google Tango visual-inertial ground truth, 200 Hz. Person-disjoint split: train hang, dan, huayi, ma, hao (59 sequences, 1.66 h); validation ruixuan, zhicheng (16, 0.49 h); test shali, tang, xiaojing, yajie (19, 0.56 h).
- Networks: 1 s window (thesis baseline), velocity + log-standard-deviation head; MSE for 10 epochs then Gaussian maximum likelihood, early stopping on validation. Training augmentation as in the thesis: random accelerometer (± 0.02 m/s²) and gyroscope (± 0.002 rad/s) biases, gravity-direction perturbation and random yaw, redrawn every epoch.
- EKF: PINNeAPLe ExtendedKalmanFilter with state (position, velocity), IMU mechanisation at 50 Hz using the phone's game rotation vector (heading aligned to the ground truth once, at the segment start), network

velocity update at 10 Hz; drift rate = final position error / distance walked, per 90 s segment.

- PINN (thesis chapter 4): network predicting the velocity over a 4 s window at 50 Hz; physics loss = mismatch between the accelerometer measurement model and the central-difference derivative of the predicted velocity; $\lambda \in \{0, 0.01, 0.1, 1\}$.
- Statistics: all comparisons are paired on the same test segments (Wilcoxon signed-rank, $\alpha = 0.05$); the bias test uses the standard error of the mean heading-frame error; test data are used once.

3 Results

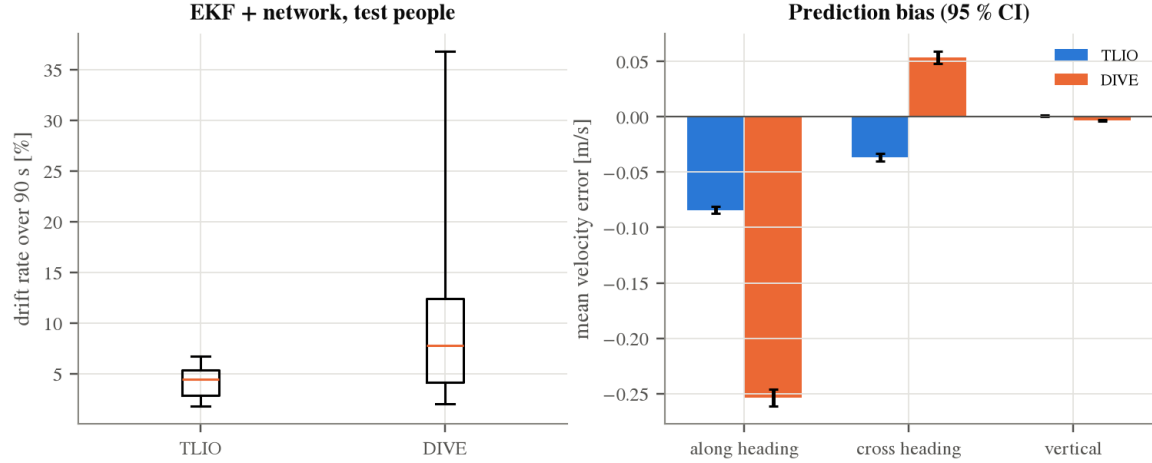


Figure 1. Left: drift over 90 s segments (whiskers 5–95 %). Right: mean velocity error in the true heading frame.

Table 1. Chapter 3 reproduction. Thesis values from its Table 3-1 (different data and drift definition).

model	standalone error median / p95 [m/s]	drift median / p95 [%]	thesis: error median / p95	thesis: drift median / p95
TLIO	0.18 / 0.45	4.5 / 6.9	0.17 / 0.84	8.1 / 28.8
DIVE	0.26 / 1.17	7.8 / 37.0	0.26 / 1.26	15.1 / 47.1
IMU only (no network)	—	69 / 147	—	—

On the same 19 test segments, DIVE's drift minus TLIO's: +3.19, $p = 0.0053$. The thesis ranking holds.

Bias of the TLIO network in the true heading frame: -0.084 m/s along, -0.036 m/s across and +0.0010 m/s vertical ($z = -51, -20, 2.5$). A consistent along-heading under-prediction is exactly the kind of non-zero-mean error that the EKF cannot absorb.

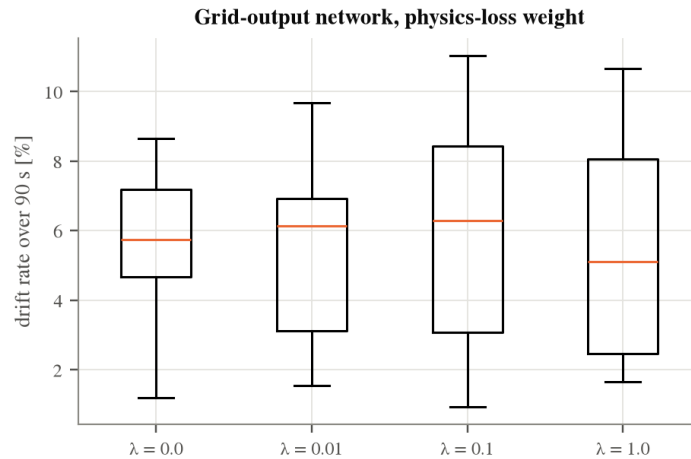


Figure 2. Chapter 4 reproduction: drift for each physics-loss weight.

Table 2. Grid-output network with physics loss.

λ	standalone error median [m/s]	drift median / p95 [%]	Δ drift vs $\lambda = 0$ (median, p)	thesis Ext. ResNet: error / drift
0.0	0.175	5.8 / 9.9	reference	0.14 / 8.9
0.01	0.170	6.1 / 10.9	-0.05, p = 0.44	0.13 / 8.3
0.1	0.183	6.3 / 12.1	-0.14, p = 0.89	0.14 / 8.2
1.0	0.217	5.1 / 12.4	+0.71, p = 0.71	0.16 / 10.9

4 Discussion

The reproduction transfers the thesis' questions to public data and checks each conclusion with a paired test rather than by comparing medians. Absolute drift rates differ from the thesis because the data, the filter and the drift definition differ; the direction of each conclusion is what can be compared. The companion paper uses the bias finding directly: if the error has a non-zero mean in the heading frame, it can be estimated on validation data and removed.

References

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